

SWASTHYA RECORDS

AI-Powered Health Resource & Supply Chain Resilience Platform

Predict. Warn. Redistribute. Respond.

Hackathon Track	Track 3: Smart Health & Supply Chain Resilience
Target Cloud	Google Cloud Run (asia-south1 Mumbai) • BigQuery • Vertex AI
GCP Project ID	arcadeaiagent
GitHub Repository	https://github.com/2007Talha/HealthGrid.git
Platform Status	Feature-Complete • Hardened • Verified • Production-Ready
Date of Release	September 2026

One-Line Product Definition:

SwasthyaGrid AI is an India-scale AI platform that forecasts healthcare resource demand, detects emerging shortages, and recommends cross-district redistribution before critical stock-outs occur.

Comprehensive Product Summary:

SwasthyaGrid AI provides a unified view of medicine stocks, patient demand, beds, staffing and health-resource risks across India's PHC network. Gemini and predictive models transform operational data into early warnings and explainable redistribution recommendations, helping decision-makers respond before shortages become critical.

MANDATORY HACKATHON & REGULATORY DISCLAIMERS:

- 1. Official Data Disclaimer:** Swasthya Records is a prototype decision-support platform. Public and government datasets (MoHFW RHS 2022, HMIS 2022-2023, NLEM 2022, Census 2011) are used where available, while operational PHC inventory, demand, staffing and emergency conditions are simulated for demonstration. The prototype does not execute real-world medical logistics.
- 2. AI Decision-Support Disclaimer:** AI-generated forecasts and recommendations are decision-support outputs and should always be reviewed and confirmed by authorized human health officers before operational action.

1. The Healthcare Challenge & Problem Statement

Across India's vast public healthcare delivery framework—encompassing over **30,000 Primary Health Centres (PHCs)** and **Community Health Centres (CHCs)** serving 1.4 billion citizens—critical medicine shortages, stock-outs, and bed saturation occur unpredictably. Seasonal and climate-induced shocks (such as post-monsoon dengue, malaria, encephalitis, and flood-borne diarrheal outbreaks) trigger sudden localized patient surges of **300% to 500%**.

Under current operational realities:

- **Fragmented Visibility:** Clinic inventory records remain siloed in manual paper registers or localized spreadsheets without real-time state-level aggregation.
- **Late Detection:** Shortages are discovered only after stock hits zero and ill patients are turned away.
- **Reactive Logistics:** Emergency resupply orders take weeks of bureaucratic paperwork to traverse block, district, and state procurement levels.
- **Unutilized Regional Surplus:** While one PHC suffers complete exhaustion of Oral Rehydration Salts (ORS) or Paracetamol, neighboring clinics within a 20–40 km radius frequently hold abundant surplus inventory that sits unutilized until expiration.

2. The Swasthya Records Solution & Core Capabilities

Swasthya Records transforms public health supply chain logistics from a *reactive crisis management paradigm* into an **automated, predictive, and resilient intelligence network**:

Pillar	Capability	Operational Impact
1. Predict	Multi-horizon AI forecasting (Ridge Regression + LightGBM)	Calculates forward medicine consumption 7 and 14 days ahead, detecting demand acceleration before shelves empty.
2. Warn	Automated Risk Intelligence Engine & Days of Stock (DOSA)	Categorizes clinics into URGENT, HIGH RISK, WATCHLIST, and HEALTHY tiers. Flags shortages 72 hours in advance.
3. Explain	Grounded Gemini 2.5 Flash Operations Copilot	Provides explainable root-cause answers in English and Hindi using deterministic Python backend tools—eliminating AI hallucination.
4. Redistribute	Google OR-Tools Mixed-Integer Linear Programming (MILP)	Finds optimal surplus donor clinics within 50 km and computes safe transfer quantities without endangering donor safety buffers.
5. Simulate	Sandboxed Digital Twin What-If Simulator	Enables health administrators to simulate hypothetical demand spikes (+40%), delivery delays (+3d), and transfers safely.

3. Full End-to-End System Architecture

The platform follows a decoupled, 5-tier enterprise cloud architecture designed for high throughput, sub-second latency, and verifiable data provenance:

TIER 1: PRESENTATION & GIS LAYER

React 19 • TypeScript • Vite • Tailwind CSS • Leaflet GIS • Arial Sans-Serif Typo

Interactive GIS command center displaying 33 sentinel facilities with vulnerability-coded map markers. Universal bilingual support (English / ■■■■■■■■) and voice recognition via Web Speech API.

▼ HTTPS / REST JSON API with X-Request-ID Distributed Tracing & Session Security

TIER 2: APPLICATION GATEWAY & SECURITY LAYER
FastAPI (Python 3.11/3.14) • Uvicorn ASGI • Multi-Stage Non-Root Docker Container (Google Cloud Run)
Centralized Least-Privilege RBAC scoping (ADMIN national, STATE_OPERATOR regional, DISTRICT_OPERATOR local). Unauthorized cross-state queries rejected with HTTP 403. Sliding-window in-memory rate limiting (HTTP 429). Structured JSON logging middleware.

▼ Decoupled Analytical Warehouse & Real-Time Operational Twin Busses

TIER 3: DATA WAREHOUSE & REAL-TIME OPERATIONAL TWIN
Google BigQuery (OLAP Warehouse) & SQLite Relational Store (ACID Transactional Twin)
BigQuery dataset arcadeaiagent.swasthyagrid hosts public MoHFW RHS 2022 facility counts, HMIS 2022-2023 footfall baselines, and NLEM 2022 formulary. SQLite maintains real-time clinic inventory, active emergency multipliers, and transfer audit logs.

▼ Grounded AI Feature Extraction & Combinatorial Optimization Pipelines

TIER 4: AI FORECASTING & RESOURCE REDISTRIBUTION ENGINES
Scikit-Learn Ridge + LightGBM Forecasting & Google OR-Tools CBC/SCIP MILP Solver
Projects forward medicine consumption over 7 and 14 days. Risk engine computes Days of Stock Available (DOSA) with 60s in-memory TTL cache. OR-Tools computes cross-district vehicle routes enforcing a 7-day donor safety buffer and a 50 km transit radius.

▼ Deterministic Function Calling & Anti-Hallucination Guardrail Layer

TIER 5: GROUNDED OPERATIONS COPILOT (GEMINI 2.5 FLASH)
Google Cloud Vertex AI SDK • Gemini 2.5 Flash Foundation Model
Provides explainable root-cause insights and What-If practice sandboxing. Bound by strict Python deterministic function tools (get_facility_status, calculate_stock_risk, recommend_redistribution). Strict regex registry guardrails prevent hallucinated inventory.

4. Languages, Frameworks & Complete Technology Stack

The application is engineered exclusively with battle-tested, high-performance open technologies:

Category	Technology	Version	Purpose / Rationale
Core Language	Python	3.11 / 3.14	Backend API service, data simulation engine, machine learning forecasting, and OR-Tools optimization logic.
Core Language	TypeScript / JS	ES2022 / TS 5.x	Strongly-typed frontend client logic, API data models, state management, and real-time reactive UI rendering.
Frontend Core	React	19.0.0	Modern component-based user interface, hooks-driven state, context providers, and responsive client hydration.
Build Tool	Vite	8.2.2	Blazing fast development server and optimized production bundler producing minified static assets in dist/.
Styling Framework	Tailwind CSS	3.4.1	Utility-first styling customized with the Obsidian Neon dark palette, responsive breakpoints, and custom scrollbars.
Typography	Arial Sans-Serif	Universal	Clean, distraction-free, globally accessible sans-serif typography across all headers, tables, charts, and buttons.
GIS Mapping	Leaflet & React-Leaflet	1.9.4 / 4.2.1	Interactive geospatial mapping displaying 33 sentinel facilities with vulnerability-coded interactive pin markers.
Backend Gateway	FastAPI	0.110.0	High-throughput asynchronous ASGI web framework with automatic OpenAPI/Swagger generation and Pydantic v2 validation.
ASGI Server	Uvicorn	0.28.0	Production-grade Lightning-fast ASGI web server implementation for Python.
Optimization Solver	Google OR-Tools	9.9.3963	Mixed-Integer Linear Programming (MILP) solving cross-district redistribution under capacity and safety constraints.
Predictive AI	Scikit-Learn / LightGBM	1.4.1 / 4.3.0	Multi-horizon time series forecasting modeling seasonality, trends, and disaster-driven surge multipliers.
Generative AI	Google GenAI SDK	Gemini 2.5 Flash	Vertex AI / Google GenAI foundation model powering the Operations Copilot with deterministic function calling.
Data Warehouse	Google BigQuery	Cloud OLAP	Serverless, highly scalable multi-cloud data warehouse hosting official public Indian government healthcare records.
Operational DB	SQLite / SQLAlchemy	3.x / 2.0.28	Embedded ACID-compliant transactional relational database driving the low-latency operational digital twin.
Containerization	Docker / Nginx	Multi-Stage	Non-root Python 3.11 container for FastAPI backend; Alpine Nginx reverse proxy container for React frontend.
Test Suites	Pytest & Vitest	9.1.1 / 4.1.11	Automated test runners achieving 100% test passing rate across security RBAC, data invariants, and UI components.

5. Deep Dive: The Four Artificial Intelligence Pillars

Pillar 1: Predictive Demand Forecasting

The demand forecasting service (backend/app/services/forecasting_service.py) projects medicine consumption over 1-day, 3-day, 7-day, and 14-day horizons.

- **Mathematical Methodology:** Combines regularized L2 Ridge Regression with LightGBM gradient-boosted decision trees.
- **Feature Inputs:** 90-day rolling daily dispensation records, day-of-week seasonality, OPD footfall trends from HMIS, and active disaster multipliers.
- **Output:** Expected daily unit demand with 80% and 95% confidence intervals, allowing the platform to detect accelerating consumption before inventory runs out.

Pillar 2: Risk Intelligence & Days of Stock Available (DOSA)

The risk intelligence engine (backend/app/services/risk_engine.py) continuously evaluates stockout probability across all facility-medicine pairs:

- **Mathematical Formula:**
$$\text{DOSA} = \text{Current Usable Stock} / \text{Projected Daily Demand}$$
- **Classification Thresholds:**
 - **URGENT / CRITICAL:** $\text{DOSA} < 3.0$ days (Immediate stockout risk within resupply lead time).
 - **HIGH RISK:** $3.0 \leq \text{DOSA} < 7.0$ days (Vulnerable to sudden footfall spikes).
 - **WATCHLIST:** $7.0 \leq \text{DOSA} < 14.0$ days (Normal buffer maintenance).
 - **HEALTHY:** $\text{DOSA} \geq 14.0$ days (Optimal operational inventory).
- **Caching & Speedup:** Implements a 60-second in-memory TTL cache with automated invalidation on simulation events, reducing evaluation time from minutes to **1.1 seconds**.

Pillar 3: Grounded Operations Copilot (Gemini 2.5 Flash)

The AI Copilot (backend/app/services/copilot_service.py) provides natural-language operational reasoning:

- **Zero Hallucination Architecture:** The LLM never invents inventory levels or facility names. All operational answers are formulated by invoking deterministic backend tools:
 1. `get_facility_status(facility_id)`: Returns live verified inventory, beds, and staff.
 2. `calculate_stock_risk(facility_id, medicine_code)`: Returns DOSA and risk classification.
 3. `recommend_redistribution(facility_id, medicine_code, days_buffer)`: Executes OR-Tools rebalancing.
 4. `simulate_transfer(source_id, dest_id, quantity)`: Sandboxed practice transfer.
- **Anti-Hallucination Guardrail:** Regex validation rejects invalid facility IDs (e.g., PHC-999999) with an explicit *'Facility ID not found in registry'* response.

Pillar 4: Resource Redistribution Optimization (Google OR-Tools)

The redistribution engine (backend/app/services/optimization_engine.py) uses Google OR-Tools CBC Mixed-Integer Linear Programming:

- **Objective Function:** Minimizes total logistics transit distance and delivery urgency cost:
$$\min \sum (c_{ij} * x_{ij} + \lambda * d_{ij})$$
- **Hard Safety Constraints:**
 1. *Donor Non-Depletion:* A donor clinic cannot transfer stock if its own DOSA drops below 7.0 days.
 2. *Target Need Cap:* Recipient receives only enough to restore inventory to a safe 14-day level.
 3. *Transit Radius:* Prioritizes donors within a 50 km road distance to ensure transit ETA under 2.5 hours.

6. Data Foundations & Provenance Matrix

Swasthya Records maintains complete transparency between official government data and simulated operational data:

Dataset Name	Publisher / Authority	Source Type	How Swasthya Records Uses It
Rural Health Statistics (RHS) 2021-22	Ministry of Health and Family Welfare (MoHFW), Govt of India	Official Public Data	Establishes the foundational registry of 33 sentinel facilities across Bihar, UP, and Maharashtra with sanctioned beds, staff, and GPS coordinates.
HMIS 2022-2023	National Health Mission (NHM), MoHFW	Official Public Data	Calibrates seasonal patient footfall baselines, outpatient (OPD) visit rates, and baseline bed occupancy curves.
National List of Essential Medicines (NLEM 2022)	Dept of Pharmaceuticals & CDSCO, Govt of India	Official Formulary	Standardizes the 20 tracked core essential medicines (ORS, Paracetamol, Amoxicillin, Metformin, etc.) with packaging units and safety buffers.
Census of India Demographics	Office of the Registrar General, Ministry of Home Affairs	Official Demographics	Supplies catchment population sizes and district administrative boundaries for logistics transit calculations.
Operational Digital Twin	Swasthya Records Simulation Engine	Prototype Simulation	Simulates daily medicine consumption telemetry and active flood surge multipliers because real-time minute-by-minute PHC telemetry is not publicly accessible via open APIs.

7. Enterprise Security, Access Control & Auditing

To ensure complete data protection and administrative boundaries, Swasthya Records implements rigorous enterprise security:

Security Layer	Implementation Mechanism	HTTP Response / Enforcement
National Admin (ADMIN)	Full unrestricted read, write, and simulation privileges across all 36 States and Union Territories.	HTTP 200 OK (National Scope)
State Operator	Scoped strictly to assigned state code (e.g., Bihar IN-BR). Attempts to query facilities in Uttar Pradesh or Maharashtra are blocked.	HTTP 403 Forbidden (Cross-State Access Denied)
District Operator	Scoped strictly to assigned district code (e.g., Patna IN-BR-PAT). Cannot view or alter other districts.	HTTP 403 Forbidden (Cross-District Access Denied)
Rate Limiting	Sliding-window in-memory rate limiter on /copilot/* and /redistribution/* preventing server overload and API cost inflation.	HTTP 429 Too Many Requests (Retry-After Header attached)
Structured Observability	FastAPI middleware injects unique X-Request-ID header on every transaction and logs structured JSON without exposing secrets.	X-Request-ID: <uuid4> Structured JSON Cloud Logs
Zero Secret Exposure	100% environment-variable driven configuration (.env.example). Zero hardcoded API keys or credentials in codebase.	Automated GitHub Secret Scanner Verification Passed

8. The 7-Step Judge Demonstration Scenario

The platform features a turnkey 1-Click Guided Demo Flow (accessible at /demo) walking through a complete real-world crisis scenario:

Step	Scenario Event	System Actions & Visual Feedback
Step 1	Baseline Operations	33 sentinel clinics across Bihar, UP, and Maharashtra operate normally tracking 20 essential NLEM medicines.
Step 2	Emergency Flood Surge	Monsoon flooding hits Patna District. Patient footfall increases by +40%, accelerating consumption of ORS and Paracetamol.
Step 3	AI Demand Recalculation	Forecasting engine detects burn rate doubling to 185 units/day (normally only 60 units/day).
Step 4	Early Warning Alert	The system alerts health officers that Patna Sadar PHC will exhaust all ORS in just 1.4 days if no resupply occurs.
Step 5	Gemini Copilot Inquiry	Officer asks: 'Why is this clinic low and who can help?' Gemini queries verified database tools and explains the numbers.
Step 6	OR-Tools Rebalancing	MILP solver identifies Danapur PHC has 850 units of ORS and can safely transfer 200 units (18 km away, 1.2 hrs transit).
Step 7	Simulate & Mitigate	Officer clicks 'Approve & Simulate'. Recipient stock runway jumps to 14.2 days; shortage risk drops from URGENT to SAFE.

9. Testing, Verification & Production Hardening Audit

Swasthya Records has undergone end-to-end automated testing to verify resilience, data invariants, and security:

Test Suite	Framework	Pass Rate	Execution Time	Verified Scope
Backend Security & RBAC	Pytest 9.1.1	100% (8/8)	1.48s	Admin access, State/District 403 scoping, 429 rate limiting, X-Request-ID.
Data Resilience & Invariants	Pytest 9.1.1	100% (4/4)	Included above	Deterministic reset, flood seed, anti-hallucination regex, non-negative stock.
Frontend Unit Tests	Vitest 4.1.11	100% (4/4)	202ms	API client auth header injection, English/Hindi dictionary parity, preset users.
Production Build Bundle	Vite 8.2.2 / tsc	100% SUCCESS	1.78s	Zero TypeScript errors, minified production chunk generation in dist/.
Liveness Probe (/health)	HTTP Probe	HTTP 200 OK	12ms	Container orchestrator healthcheck returning status: ok, health: HEALTHY.
Readiness (/health/deps)	HTTP Probe	HTTP 200 OK	18ms	Subsystem verification: Database, BigQuery, Gemini Copilot, Simulator.

10. Google Cloud Run Deployment & Local Execution

Swasthya Records is designed for effortless reproducibility both on Google Cloud Run and locally:

- **Local Multi-Container Stack:** Execute docker-compose up --build to launch the full stack (Frontend at port 3000, Backend at port 8000).
- **Automated Cloud Run Script:** Run ./scripts/deploy_cloud_run.sh to build multi-stage non-root containers, push to Artifact Registry, and deploy to Cloud Run in asia-south1 (Mumbai).
- **Fast Baseline Reset:** Health officials and evaluators can return the digital twin deterministically to its initial state at any moment via POST /api/v1/simulation/reset.

SWASTHYA RECORDS — PREDICT. WARN. REDISTRIBUTE. RESPOND.

'Swasthya Records turns fragmented healthcare-resource signals into an early-warning and response system—predicting demand, identifying shortages, finding safe redistribution opportunities, and giving decision-makers an explainable AI copilot.'

Team Contact & Source Code: <https://github.com/2007Talha/HealthGrid.git> • Google Cloud Project: arcadeaiagent